

A KERNEL-CHECKED SYMMETRIC-SQUARE STRUCTURE THEOREM FOR COOPER’S SPORADIC APÉRY-LIKE OPERATORS, AND THE MONODROMY LATTICE OF THE s_7 FAMILY

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ABSTRACT. Cooper’s three sporadic Apéry-like sequences s_7, s_{10}, s_{18} satisfy order-3 recurrences belonging to a four-parameter template. We prove — uniformly in the template parameters $(a, b, c, d) \in \mathbb{Q}^4$, with the proof verified by the Lean 4 kernel — that the associated order-3 operator L_3 factors as $P_2 \cdot \text{Sym}^2(L_2)$ for an order-2 partner operator L_2 whose coefficients are determined explicitly by the template, and, independently, that the Almkvist–van Straten symmetric-square criterion $W \equiv 0$ holds identically on the template. The partner separates the three sporadic candidates arithmetically: only the s_7 partner has an integral holomorphic solution (OEIS A279619), and we make precise the mechanism — a *normalized* integral uniformizer — underlying that integrality. We further prove, exactly over $\mathbb{Q}$, that L_2 and L_3 are irreducible and that L_3 is the minimal-order annihilator of its holomorphic solution. Finally, we compute the monodromy of the s_7 family numerically to 60 digits, recognize the entries exactly (largest residual $\approx 10^{-59}$), and derive — by an exact integer pipeline verified by two independently written implementations — that the joint monodromy-invariant lattice is isometric over $\mathbb{Z}$ to $U \oplus \langle 14 \rangle$, in exact agreement with the lattice $(M_7)^\perp$ of Dolgachev’s M_n -polarized K3 framework. We are explicit throughout about epistemic status: which statements are kernel-checked, which are exact symbolic computation, which are numerically supported, and which are classical results cited. All computations are reproducible from the accompanying repositories.

CONTENTS

1.	Introduction	3
1.1.	Results	3
1.2.	What is new	5
1.3.	Organization	5
2.	Preliminaries and notation	5
2.1.	Operators	5
2.2.	Riemann schemes and Fuchsian invariants	6
2.3.	Cooper’s template and the three sporadic sequences	6
2.4.	Lattices	6
3.	The two operators and their exact Riemann schemes	7

3.1. The order-3 operator	7
3.2. The order-2 partner	7
4. The template structure theorem	8
4.1. The symmetric square in θ -form	8
4.2. The partner is determined, not chosen	8
4.3. The structure theorem	10
4.4. The three sporadic instances	10
4.5. The Almkvist–van Straten criterion on the template	11
5. Irreducibility and minimality	12
5.1. No hyperexponential solution	12
5.2. The monodromy is not dihedral	13
5.3. Irreducibility of L_3	14
6. The monodromy lattice of the s_7 family	15
Disclosure	15
6.1. Numerical continuation	16
6.2. The exact lattice pipeline	16
6.3. The integral splitting	17
6.4. The s_{10} control	18
7. The arithmetic separation of the three candidates	19
7.1. The partner recurrence	19
7.2. Non-integrality for s_{10} and s_{18}	20
7.3. Integrality for s_7	20
7.4. The mechanism: a normalized integral uniformizer	21
7.5. The modular coordinate	22
8. Relation to the M_n -polarized framework	23
8.1. The framework, as cited	23
8.2. The logical direction	23
8.3. The evidence, assembled	24
8.4. A conditional rank statement	24
8.5. What is conjectural	25
8.6. What is not claimed	26
8.7. Open questions	26
9. The formal development	26
9.1. Toolchain and layout	27
9.2. The results, and how they are stated	27
9.3. The axiom inventory, in full	28
9.4. Golden validation and $\text{PASS}(N)$ discipline	29
9.5. What the kernel did not check	29
9.6. An honest assessment of the formalization	29
10. Reproducibility	30
10.1. Artifact index	30
10.2. Running the artifacts	31
10.3. Negative controls	31
References	32

1. INTRODUCTION

Apéry's proof of the irrationality of $\zeta(3)$ placed a handful of integer sequences — defined by three-term recurrences with polynomial coefficients, yet integral against all apparent odds — at the center of a circle of ideas connecting holonomic differential equations, modular forms, and the geometry of families of algebraic surfaces. The six *sporadic* Apéry-like sequences of order 2, catalogued by Zagier, and their order-3 companions have since been objects of sustained experimental and theoretical attention. Cooper [3] identified three further sporadic sequences, commonly denoted s_7 , s_{10} and s_{18} , satisfying order-3 recurrences; Gorodetsky [7] gave closed forms and placed all three inside a single four-parameter recurrence template. Almkvist and van Straten [1] exhibited explicit projective K3 models whose periods realize the corresponding order-3 operators.

This paper studies the differential operators attached to Cooper's template from three angles — structural, arithmetic, and lattice-theoretic — with a methodological commitment that we regard as part of the contribution: *every claim is labeled by how it is known*. Specifically, each statement in this paper carries exactly one of the following statuses, and the label is part of the statement's presentation, not a footnote:

- (**K**) *Kernel-checked*: proved in Lean 4 and verified by its proof kernel against the pinned `mathlib` library [4, 12, 13], with the axiom base disclosed in §9;
- (**E**) *Exact*: established by exact symbolic computation over $\mathbb{Q}$ (no floating point), reproducible by a named checker script in the accompanying repositories, each shipped with negative controls (§10);
- (**N**) *Numerically supported*: derived from high-precision numerics followed by exact recognition and exact post-verification; stated as a *Numerical Claim*, never as a theorem;
- (**C**) *Cited*: a classical result of the literature, used as stated, with the precise statement and location given.

1.1. **Results.** Write $\theta = z \frac{d}{dz}$. Cooper's template, in the normalization of [7], is the family of order-3 operators

$$(1) \quad L_3(a, b, c, d) = \theta^3 - z(2\theta + 1)(a\theta^2 + a\theta + b) + z^2(c(\theta + 1)^3 + d(\theta + 1)), \quad (a, b, c, d) \in \mathbb{Q}^4,$$

with $(a, b, c, d) = (13, 4, -27, 3)$, $(6, 2, -64, 4)$, $(14, 6, 192, -12)$ for s_7 , s_{10} , s_{18} respectively [3, 7].

Our first result is a uniform structure theorem (Theorem 4.5, status (**K**)): for *every* parameter choice, $L_3(a, b, c, d)$ is, in a division-free θ -form sense made precise in §4, the product $P_2 \cdot \text{Sym}^2(L_2)$ of its own leading θ -coefficient with the symmetric square of an order-2 operator L_2 whose coefficients are *determined* by the template — the partner is solved out of the identities, leaving a single nontrivial constraint, which holds identically. Independently, the Almkvist–van Straten symmetric-square criterion $W \equiv 0$ holds identically

on the template in cleared-denominator form (Theorem 4.9, status **(K)**). Both facts are single kernel-checked proofs covering s_7 , s_{10} , s_{18} — and any future instance of the template — simultaneously by specialization. To our knowledge this is the first kernel-checked verification of a symmetric-square structure theorem for Picard–Fuchs-type operators in any proof assistant.

The partner operator makes an arithmetic distinction visible (§7): among Cooper’s three sporadic sequences, only the s_7 partner has an integral holomorphic solution — the sequence 1, 2, 22, 336, 6006, . . . (OEIS A279619 [11]). Non-integrality for s_{10} and s_{18} is kernel-checked by finite witness (status **(K)**); integrality for s_7 is a theorem of O’Brien [10] (status **(C)**), whose applicability to our partner is kernel-checked at the level of recurrences. We isolate the mechanism: the modular coordinate $X_7 = \eta_1^3 \eta_7^3 / z_7^3$ is a *normalized* integral uniformizer, and normalization — not mere integrality of η -quotients — is the load-bearing property (status **(E)**).

Third (§5, status **(E)**): the s_7 and s_{10} partners L_2 are irreducible, by a denominator obstruction — every attainable residue sum of a hyperexponential logarithmic derivative lies in $\frac{1}{2}\mathbb{Z}$, while the exponents of L_2 at infinity are $\{\frac{1}{3}, \frac{2}{3}\}$ (resp. $\{\frac{3}{8}, \frac{5}{8}\}$) — and their monodromy is not dihedral, because the doubled indicial root at 0 forces a logarithmic solution and no nontrivial unipotent element lies in the normalizer of a maximal torus of SL_2 . Consequently $L_3 = \mathrm{Sym}^2(L_2)$ is irreducible and is the *minimal-order* annihilator of its holomorphic solution. All steps are exact over $\mathbb{Q}$.

Finally (§6), we compute the monodromy representation of the s_7 partner by numerical analytic continuation at 60-digit working precision and recognize the entries exactly; the largest recognition residual is $\approx 10^{-59}$ against an acceptance gate of 10^{-35} , and the recognized matrices pass exact structural post-verification (involutivity, the cusp relation at infinity, existence and uniqueness up to scale of a joint invariant form, orbit-lattice closure). This step is numerics-backed and is reported as Numerical Claim 6.2 (status **(N)**), not as a theorem. Taking the recognized matrices as input, an exact integer-arithmetic pipeline then derives the joint monodromy-invariant lattice in its primitive even scaling and proves — unconditionally, as a statement about the exhibited Gram matrix (Proposition 6.4, status **(E)**) — that it is isometric over $\mathbb{Z}$ to $U \oplus \langle 14 \rangle$: Gram matrix $\begin{bmatrix} 0 & 0 & -1 \\ 0 & 14 & 0 \\ -1 & 0 & 0 \end{bmatrix}$, determinant -14 , signature $(2, 1)$, discriminant form $\mathbb{Z}/14$ with $q = \frac{1}{14}$. The isometry is realized by an explicit $\mathrm{GL}_3(\mathbb{Z})$ base change and has been verified by *two independently written* exact implementations; the identical pipeline applied to the s_{10} family derives determinant -20 and $U \oplus \langle 20 \rangle$, so the computation discriminates between levels. This matches on the nose the lattice $(M_7)^\perp = U \oplus \langle 14 \rangle$ of Dolgachev’s M_n -polarized K3 framework [5], in which the Picard–Fuchs operator is a symmetric square by a theorem of Doran [6]; §8 assembles the evidence that the s_7 family is M_7 -polarized, and states precisely what remains unproven — in particular, the identification of the computed monodromy

lattice with the transcendental lattice of the family is presented conditionally, not as a theorem.

1.2. What is new. None of the framework results are ours: symmetric-square structure for M_n -polarized families is Doran's theorem, and $(M_n)^\perp = U \oplus \langle 2n \rangle$ is Dolgachev's computation. The contributions of this paper are: (i) the uniform, kernel-checked template theorem with an explicitly determined partner, where the literature treats candidates one at a time and asserts existence rather than exhibiting the partner from the parameters; (ii) the exact irreducibility/minimality proofs for these specific operators; (iii) the sharpened integrality mechanism for s_7 , including the checked resolution of an ambiguity in the source's defining equation; (iv) the explicit monodromy lattice of the s_7 family with an exact, doubly-verified integral splitting $U \oplus \langle 14 \rangle$ — a *converse-direction* computation: rather than deducing invariants from a polarization assumption, we compute the invariants of a concretely given operator family and match them against the framework, in a situation where, as Doran notes ([6], §6), the classification of rank-19 lattice polarizations is not complete; and (v) the epistemic architecture itself — kernel-checked/exact/numerical/cited labeling with reproducible artifacts and negative controls — which we offer as a methodological contribution to experimental mathematics.

1.3. Organization. §2 fixes notation. §3 presents the two operators for s_7 and their exact Riemann schemes. §4 states and proves the template structure theorem. §5 proves irreducibility and minimality. §6 presents the monodromy computation and the integral splitting. §7 treats the s_7 integrality mechanism. §8 relates the results to the Dolgachev–Doran framework and states the conditional statements and open questions. §9 describes the Lean 4 formalization, including its complete axiom inventory. §10 indexes the artifacts needed to reproduce every number in this paper.

2. PRELIMINARIES AND NOTATION

2.1. Operators. Throughout, z is the affine coordinate, $D = \frac{d}{dz}$, and $\theta = zD$ is the Euler operator. We work with linear differential operators with coefficients in $\mathbb{Q}[z]$ (occasionally $\mathbb{Q}(z)$ when a monic normalization is needed), acting on formal power series and on multivalued analytic functions on $\mathbb{P}^1$ minus the singular locus.

An order-2 operator will be written in θ -form as

$$(2) \quad L_2 = P_2 \theta^2 + P_1 \theta + P_0, \quad P_i \in \mathbb{Q}[z],$$

and an order-3 operator as $L_3 = c_3 \theta^3 + c_2 \theta^2 + c_1 \theta + c_0$ with $c_i \in \mathbb{Q}[z]$. For a monic operator in D -form, $L = D^2 + pD + q$ with p, q rational functions, the *symmetric square* $\text{Sym}^2(L)$ is the unique monic order-3 operator annihilating all products $y_i y_j$ of solutions of L ; explicitly

$$(3) \quad \text{Sym}^2(D^2 + pD + q) = D^3 + 3p D^2 + (2p^2 + p' + 4q)D + (4pq + 2q').$$

Formula (3) is validated in the formalization against two examples derived from first principles (§9): the harmonic oscillator, $\text{Sym}^2(D^2 + 1) = D^3 + 4D$ (solution products spanned by $1, \cos 2t, \sin 2t$), and $\text{Sym}^2(D^2 + D) = D^3 + 3D^2 + 2D$ (solution products $1, e^{-t}, e^{-2t}$).

2.2. Riemann schemes and Fuchsian invariants. For a Fuchsian operator L of order n on $\mathbb{P}^1$, the *Riemann scheme* records the local exponents of L at each singular point. The Fuchs relation states that the sum of all exponents, over all singular points, equals $\binom{n}{2}(s-2)$ where s is the number of singular points; for an order-3 operator with four singular points the required total is 6. A point of *maximal unipotent monodromy* (MUM) is one whose exponents all coincide (here: $0, 0, 0$ at $z = 0$), so that the local monodromy is a single Jordan block. We write $W(L)$ for the Wronskian of a fundamental system of solutions of L .

2.3. Cooper's template and the three sporadic sequences. Cooper [3] introduced three sequences s_7, s_{10}, s_{18} satisfying order-3 Apéry-like recurrences; Gorodetsky [7] subsumed them in the template (1), whose θ -form coefficients, after expanding and moving powers of z to the left (the inner factors have constant coefficients, so no commutator terms arise), are

$$(4) \quad \begin{aligned} c_3 &= 1 - 2az + cz^2, & c_2 &= -3az + 3cz^2, \\ c_1 &= -(a + 2b)z + (3c + d)z^2, & c_0 &= -bz + (c + d)z^2. \end{aligned}$$

The holomorphic solution of $L_3(a, b, c, d)$ at $z = 0$ normalized by $u(0) = 1$ has coefficients satisfying Cooper's three-term recurrence; for $(a, b, c, d) = (13, 4, -27, 3)$ this is s_7 (OEIS A183204: $1, 4, 48, 760, \dots$), for $(6, 2, -64, 4)$ it is s_{10} (OEIS A005260, the sums $\sum_k \binom{n}{k}^4$), and for $(14, 6, 192, -12)$ it is s_{18} [3, 7]. The parameter values are transcribed from the literature and are pinned, with source citations attached to the definitions, in the formal development (§9).

2.4. Lattices. A *lattice* is a free $\mathbb{Z}$ -module of finite rank with a nondegenerate symmetric bilinear form; it is *even* if $\langle x, x \rangle \in 2\mathbb{Z}$ for all x . U denotes the hyperbolic plane $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $\langle m \rangle$ the rank-1 lattice with Gram matrix (m) . The *discriminant group* of a lattice T is $T^\vee/T$, carrying its discriminant quadratic form $q: T^\vee/T \rightarrow \mathbb{Q}/2\mathbb{Z}$. Two lattices are isometric over $\mathbb{Z}$ if their Gram matrices are conjugate under $\text{GL}_n(\mathbb{Z})$.

For a K3 surface X , $\text{NS}(X) \subset H^2(X, \mathbb{Z})$ is the Néron–Severi lattice, of rank $\rho(X)$, and the *transcendental lattice* $T(X)$ is its orthogonal complement, of rank $22 - \rho(X)$. Following Dolgachev [5], for M a fixed even lattice of signature $(1, t)$, an *M-polarized* K3 surface is one with a primitive lattice embedding $M \hookrightarrow \text{NS}(X)$; for the rank-19 lattices $M_n = U \oplus E_8^2 \oplus \langle -2n \rangle$, Dolgachev computes $(M_n)^\perp = U \oplus \langle 2n \rangle$ ([5], §7) and proves that the coarse moduli space of M_n -polarized K3 surfaces is the Fricke modular curve $H/\Gamma_0(n) +$ ([5], Thm 7.1).

3. THE TWO OPERATORS AND THEIR EXACT RIEMANN SCHEMES

We now specialize to s_7 ; the s_{10} data are recorded in parallel where they illuminate the structure. All results in this section have status **(E)**: they are exact computations over $\mathbb{Q}$, reproducible by the checkers named in §10.

3.1. The order-3 operator. For $(a, b, c, d) = (13, 4, -27, 3)$, the leading θ -coefficient of $L_3 = L_3(13, 4, -27, 3)$ is

$$c_3 = 1 - 26z - 27z^2 = -(z + 1)(27z - 1),$$

so the finite singular points are $z = -1$ and $z = \frac{1}{27}$, together with $z = 0$ and $z = \infty$. (For s_{10} : $c_3 = 1 - 12z - 64z^2 = -(4z + 1)(16z - 1)$, singular at $-\frac{1}{4}$ and $\frac{1}{16}$.)

Proposition 3.1 (Riemann scheme of L_3 ; status (E)). *The operator L_3 is Fuchsian with Riemann scheme*

$$\left(\begin{array}{cccc} 0 & -1 & \frac{1}{27} & \infty \\ 0 & 0 & 0 & \frac{2}{3} \\ 0 & \frac{1}{2} & \frac{1}{2} & 1 \\ 0 & 1 & 1 & \frac{4}{3} \end{array} \right)$$

The exponent sum is 6, as the Fuchs relation requires for an order-3 operator with four singular points; $z = 0$ is a point of maximal unipotent monodromy; and the Wronskians satisfy $W(L_3) = W(L_2)^3$ for the partner operator L_2 of §4.

For s_{10} the scheme is the same at 0 and at the two finite points $(-\frac{1}{4}, \frac{1}{16})$, exponents $0, \frac{1}{2}, 1$, with ∞ -exponents $\{\frac{3}{4}, 1, \frac{5}{4}\}$ and Fuchs sum again 6.

3.2. The order-2 partner. The partner operator produced by Theorem 4.5 below is, for s_7 , in the θ -form (2):

$$(5) \quad P_2 = 1 - 26z - 27z^2, \quad P_1 = -13z - 27z^2, \quad P_0 = -2z - 6z^2.$$

Proposition 3.2 (Riemann scheme of L_2 ; status (E)). *The operator L_2 with coefficients (5) is Fuchsian with Riemann scheme*

$$\left(\begin{array}{cccc} 0 & -1 & \frac{1}{27} & \infty \\ 0 & 0 & 0 & \frac{1}{3} \\ 0 & \frac{1}{2} & \frac{1}{2} & \frac{2}{3} \end{array} \right)$$

In particular the local exponent differences are 0 at the MUM point $z = 0$, $\frac{1}{2}$ at both finite singular points, and $\frac{1}{3}$ at infinity. The exponents of Proposition 3.1 are exactly the pairwise sums of these, as they must be for a symmetric square. (For the s_{10} partner: ∞ -exponents $\{\frac{3}{8}, \frac{5}{8}\}$, finite exponents as for s_7 .)

Remark 3.3 (Implied Fuchsian signature; status (E) for the signature, supporting evidence for the identification). Reading the exponent differences of Proposition 3.2 as angles of a hyperbolic triangle group datum gives genus 0, elliptic points of orders $(2, 2, 3)$, one cusp, and hyperbolic area $2\pi \cdot \frac{2}{3}$. This

is precisely the signature of the Fricke group $\Gamma_0(7)+$: $\Gamma_0(7)$ has index 8 in $\mathrm{PSL}_2(\mathbb{Z})$ (area $2\pi \cdot \frac{4}{3}$), the Fricke involution halves the area, fuses the two cusps into one and the two order-3 points into one, and its two fixed points (class numbers $h(-7) = h(-28) = 1$) become the two order-2 elliptic points. The identification of the modular coordinate with a $\Gamma_0(7)+$ Hauptmodul is taken up in §7. The two finite singular points are thus order-2 elliptic points of the quotient curve — *not* degeneration points of some elliptic fibration; no such reading is made anywhere in this paper.

4. THE TEMPLATE STRUCTURE THEOREM

This section proves that *every* operator of Cooper’s template (1) is the symmetric square of an order-2 operator whose coefficients are determined by the template parameters, and that the Almkvist–van Straten self-adjointness criterion holds identically on the template. Both statements have status **(K)**: they are verified by the Lean 4 kernel, uniformly in $(a, b, c, d) \in \mathbb{Q}^4$, so that s_7, s_{10}, s_{18} — and any further instance of the template — are specializations of a single proof rather than three separate computations.

4.1. The symmetric square in θ -form. The symmetric-square formula (3) was stated for the derivation $D = \frac{d}{dz}$. We use it for $\theta = zD$ instead. This is legitimate and requires no re-derivation:

Lemma 4.1 (Change of derivation). *Let δ be any derivation of a differential field K of characteristic 0 and let $K[\delta]$ be the associated skew polynomial ring, so that $\delta f = f\delta + \delta(f)$ for $f \in K$. For $L = \delta^2 + \alpha\delta + \beta$ with $\alpha, \beta \in K$, the monic order-3 operator annihilating all products of solutions of L is*

$$(6) \quad \mathrm{Sym}^2(L) = \delta^3 + 3\alpha\delta^2 + (2\alpha^2 + \delta(\alpha) + 4\beta)\delta + (4\alpha\beta + 2\delta(\beta)).$$

Proof. The classical derivation of (3) — set $u = y_i y_j$, differentiate four times, and eliminate y'_i, y'_j using $\delta^2 y = -\alpha\delta y - \beta y$ — uses only the commutation rule $\delta f = f\delta + \delta(f)$ and the field operations. It is therefore valid verbatim for any derivation, in particular for $\delta = \theta$ on $K = \mathbb{Q}(z)$, where $\theta(f) = zf'$. $\square$

Throughout this section $\delta = \theta$, and we write $\theta(f)$ for the action on a coefficient (as opposed to θ as an operator symbol). On polynomials, $\theta(f) = zf'$; this is exactly the map `thetaC` of the formalization.

4.2. The partner is determined, not chosen. Write an order-2 operator in the θ -form (2), $L_2 = P_2\theta^2 + P_1\theta + P_0$, and suppose we ask for

$$(7) \quad L_3(a, b, c, d) = P_2 \cdot \mathrm{Sym}^2\left(\frac{1}{P_2}L_2\right),$$

an identity of operators with coefficients in $\mathbb{Q}(z)$. Comparing θ -coefficients of (7) against the template coefficients (4), and using (6) with $\alpha = P_1/P_2$,

$\beta = P_0/P_2$, gives four conditions. Read in order, the first three *solve* for the partner and the fourth is then a constraint carrying all the content:

$$(8) \quad \underbrace{c_3 = P_2}_{\text{fixes } P_2}, \quad \underbrace{c_2 = 3P_1}_{\text{fixes } P_1}, \quad \underbrace{c_1 = \theta(P_1) + 4P_0}_{\text{fixes } P_0}, \quad \underbrace{c_0 = 2\theta(P_0)}_{\text{a constraint}}.$$

Solving the first three against (4) yields the candidate partner

$$(9) \quad P_2 = 1 - 2az + cz^2, \quad P_1 = -az + cz^2, \quad P_0 = -\frac{b}{2}z + \frac{c+d}{4}z^2.$$

Nothing is free at this point. We emphasise that (9) is a *closed-form* partner exhibited from the template parameters; no geometric hypothesis and no candidate-by-candidate search enters.

Remark 4.2 (The partner leaves $\mathbb{Z}[z]$). The halved and quartered coefficients in (9) are genuine: the partner lives in $\mathbb{Q}[z]$, not $\mathbb{Z}[z]$, even for integral template parameters. For s_7, s_{10}, s_{18} the denominators happen to clear — $P_0 = -2z - 6z^2, -z - 15z^2, -3z + 45z^2$ respectively — but that is arithmetic accident per candidate, not a property of the template.

Two identities make (7) work, and both hold identically in (a, b, c, d) .

Lemma 4.3 (Structural collapse of the θ^2 -coefficient; status (K)). *For every $(a, b, c, d) \in \mathbb{Q}^4$, $2c_2 = 3\theta(c_3)$.*

Lemma 4.4 (The magic collapse; status (K)). *For every $(a, b, c, d) \in \mathbb{Q}^4$, the partner (9) satisfies $\theta(P_2) = 2P_1$.*

Proof of Lemmas 4.3 and 4.4. Both are polynomial identities in $\mathbb{Q}[z]$ with parameters: $\theta(1 - 2az + cz^2) = -2az + 2cz^2 = 2(-az + cz^2)$, and $2(-3az + 3cz^2) = 3(-2az + 2cz^2)$. In the formalization they are `partner_magic` and `cooperC2_eq_thetaC_cooperC3`, each closed by `ring` after pushing `Polynomial.derivative` through the algebraic structure, so the differentiation itself is kernel-computed rather than transcribed. $\square$

Lemma 4.3 explains why the second condition of (8) is not an independent assumption: once P_2 is matched to c_3 , requiring the θ^2 -residual to vanish *forces* $\theta(P_2) = 2P_1$. Lemma 4.4 is what makes the identity division-free, as follows. Let D_k denote the residual between the θ^k -coefficient of $\frac{1}{c_3}L_3$ and that of $\text{Sym}^2(\frac{1}{P_2}L_2)$. Clearing denominators produces the terms $P_1\theta(P_2)$ and $2P_0\theta(P_2)$; substituting $\theta(P_2) = 2P_1$ turns them into $2P_1^2$ and $4P_0P_1$, which cancel against the $-2P_1^2$ and $-4P_1P_0$ already present. Every surviving term then carries exactly one factor of P_2 , so

$$(10) \quad P_2^2 D_2 = P_2 (c_2 - 3P_1), \quad P_2^2 D_1 = P_2 (c_1 - \theta(P_1) - 4P_0), \quad P_2^2 D_0 = P_2 (c_0 - 2\theta(P_0)),$$

and the rational-function identity (7), which a priori needs P_2^2 in denominators, reduces to three polynomial identities in $\mathbb{Q}[z]$.

4.3. The structure theorem.

Theorem 4.5 (Cooper-template structure theorem; status (K)). *Let $(a, b, c, d) \in \mathbb{Q}^4$ and let $P_2, P_1, P_0 \in \mathbb{Q}[z]$ be as in (9). Then all four conditions of (8) hold identically:*

$$c_3 = P_2, \quad c_2 = 3P_1, \quad c_1 = \theta(P_1) + 4P_0, \quad c_0 = 2\theta(P_0).$$

Consequently, on the open set where $P_2 \neq 0$,

$$L_3(a, b, c, d) = P_2 \cdot \text{Sym}^2\left(\theta^2 + \frac{P_1}{P_2}\theta + \frac{P_0}{P_2}\right),$$

so every operator of Cooper's template is the leading θ -coefficient of itself times the symmetric square of an explicitly determined order-2 operator.

Proof. The first identity is a definitional equality (`partner_res3`: the polynomial $1 - 2az + cz^2$ is literally both c_3 and P_2). The second is Lemma 4.3 combined with (9) (`partner_res2`). For the third, $\theta(P_1) = -az + 2cz^2$ and $4P_0 = -2bz + (c + d)z^2$, whose sum is $-(a + 2b)z + (3c + d)z^2 = c_1$ (`partner_res1`). For the fourth, $2\theta(P_0) = -bz + (c + d)z^2 = c_0$ (`partner_res0`). Only the fourth carries content: P_2, P_1, P_0 are already pinned by the first three, so the equation either holds or the template is not a symmetric square. It holds, identically in (a, b, c, d) .

The displayed factorization then follows from (10): each residual D_k satisfies $P_2^2 D_k = P_2 \cdot 0 = 0$, and $P_2 \neq 0$ in the field $\mathbb{Q}(z)$, so $D_2 = D_1 = D_0 = 0$.

In the formalization the four identities are `partner_res3`, `partner_res2`, `partner_res1`, `partner_res0` of `Agora/Sequences/PartnerOperators.lean`, each proved once for a universally quantified parameter record. Two of them require relating the halved and quartered coefficients of P_0 to their cleared forms by hand (the constant-embedding C is opaque to `ring`); this is done by the auxiliary lemmas `C_half_b` and `C_quarter_cd`. $\square$

Remark 4.6 (Where the theorem lives, and where it does not). Theorem 4.5 is an identity in $\mathbb{Q}[z]$ and in $\mathbb{Q}(z)[\theta]$. It asserts nothing about modularity, integrality, or geometry of the partner — those are the subjects of §7 and §8, and they separate the three sporadic candidates sharply even though the structure theorem does not. Keeping “has a symmetric-square partner” distinct from “has a modular companion” is the reason the theorem is stated at template level.

4.4. The three sporadic instances.

Corollary 4.7 (status (K)). *Specializing Theorem 4.5 at the parameter values of §2 gives the order-2 partners*

$$\begin{aligned} s_7 : \quad & P_2 = 1 - 26z - 27z^2, \quad P_1 = -13z - 27z^2, \quad P_0 = -2z - 6z^2, \\ s_{10} : \quad & P_2 = 1 - 12z - 64z^2, \quad P_1 = -6z - 64z^2, \quad P_0 = -z - 15z^2, \\ s_{18} : \quad & P_2 = 1 - 28z + 192z^2, \quad P_1 = -14z + 192z^2, \quad P_0 = -3z + 45z^2, \end{aligned}$$

with finite singular points $\{-1, \frac{1}{27}\}$, $\{-\frac{1}{4}, \frac{1}{16}\}$ and $\{\frac{1}{12}, \frac{1}{16}\}$ respectively.

The s_7 and s_{10} partners had been obtained in the program by an independent computer-algebra extraction before the template argument was found; that the generic construction reproduces them exactly is a genuine cross-check of (9) rather than a restatement, and it is recorded as such in the formalization (`s7_P2_eq` and its siblings). For s_{18} there was no independent extraction to compare against: its partner is obtained here with no new algebra and no literature transcription, which matters because no order-2 companion for s_{18} appears to have been identified previously.

Remark 4.8 (Not an instance of the order-2 sporadic template; status (K)). The partner operators are *not* instances of the classical order-2 Apéry-like template, whose θ^1 -coefficient is θ of its θ^2 -coefficient. The partners instead satisfy $P_1 = \theta(P_2)/2$ (Lemma 4.4); the two templates agree only when $\theta(P_2) = 0$. For s_7 , matching the θ^2 -coefficient would force θ^1 -coefficient $-26z - 54z^2$, whereas the actual P_1 is $-13z - 27z^2$. This is recorded as a theorem (`s7_P1_ne_thetaC_P2`, `s10_P1_ne_thetaC_P2`) so that the partner is not re-derived by instantiating the wrong template.

4.5. The Almkvist–van Straten criterion on the template. Independently of Theorem 4.5, the template satisfies the Almkvist–van Straten self-adjointness criterion for an order-3 operator to be a symmetric square [1]. Writing L_3 in D -form as $\pi_3 D^3 + \pi_2 D^2 + \pi_1 D + \pi_0$ and setting $a_i = \pi_i/\pi_3$, the criterion is the vanishing of

$$(11) \quad W = \frac{1}{3}a_2'' + \frac{2}{3}a_2a_2' + \frac{4}{27}a_2^3 + 2a_0 - \frac{2}{3}a_1a_2 - a_1'.$$

The $\theta \rightarrow D$ conversion $\theta = zD$, $\theta^2 = z^2D^2 + zD$, $\theta^3 = z^3D^3 + 3z^2D^2 + zD$ applied to (1) gives

$$(12) \quad \begin{aligned} \pi_3 &= z^3(1 - 2az + cz^2), & \pi_2 &= 3z^2(1 - 3az + 2cz^2), \\ \pi_1 &= z(1 - (6a + 2b)z + (7c + d)z^2), & \pi_0 &= z(-b + (c + d)z). \end{aligned}$$

Theorem 4.9 ($W \equiv 0$ on the template; status (K)). Define $\mathcal{P} \in \mathbb{Q}[z]$ to be the numerator obtained by substituting $a_i = \pi_i/\pi_3$ into (11) and clearing denominators, i.e. $\mathcal{P} = 27\pi_3^3 W$, expanded as the six-term polynomial expression

$$\begin{aligned} \mathcal{P} &= 9(\pi_2''\pi_3^2 - \pi_2\pi_3\pi_3'' - 2\pi_2'\pi_3\pi_3' + 2\pi_2(\pi_3')^2) + 18\pi_2(\pi_2'\pi_3 - \pi_2\pi_3') + 4\pi_2^3 \\ &\quad + 54\pi_0\pi_3^2 - 18\pi_1\pi_2\pi_3 - 27\pi_3(\pi_1'\pi_3 - \pi_1\pi_3'). \end{aligned}$$

Then $\mathcal{P} = 0$ in $\mathbb{Q}[z]$ for every $(a, b, c, d) \in \mathbb{Q}^4$. Since π_3 is a nonzero polynomial for every parameter choice (its lowest coefficient is that of z^3 , equal to 1), this is equivalent to $W \equiv 0$.

Proof. After expanding (12) and computing every derivative formally, $\mathcal{P}$ is a polynomial in z whose coefficients are polynomials in (a, b, c, d) ; all of them vanish. The formalization is `P_cleared_eq_zero` in `Agora/Swampland/SymSquareC3b.lean`, closed by `ring` after `simp` pushes `Polynomial.derivative` and the constant embedding through the algebraic structure. The derivatives are genuine `Polynomial.derivative` applications rather than hand-transcribed

closed forms, so a transcription error in a differentiation cannot slip past ring. The three sporadic cases are one-line specializations (`P_cleared_s7`, `P_cleared_s10`, `P_cleared_s18`). $\square$

Remark 4.10 (Two routes, one conclusion). Theorems 4.5 and 4.9 are logically independent verifications of the same structural fact via different mechanisms: the first exhibits the partner and checks a factorization; the second checks a differential invariant that does not mention the partner at all. The cleared form $\mathcal{P}$ was, in the program’s workflow, derived twice by independent computer-algebra routes and required to agree before being encoded in Lean.

Remark 4.11 (Neither theorem discriminates among candidates). Both statements hold for *all* (a, b, c, d) . Symmetric-square structure is therefore an entry ticket for the whole template and cannot, by itself, distinguish s_7 from s_{10} or s_{18} . What does distinguish them is arithmetic: the integrality of the partner’s holomorphic solution (§7).

Remark 4.12 (Validation of the Sym^2 formula; status (K)). Formula (3) is not taken on trust in the formalization. It is validated by two golden tests against examples whose symmetric squares are known from first principles: $\text{Sym}^2(D^2 + 1) = D^3 + 4D$ (solutions $\sin t, \cos t$; products spanned by $1, \cos 2t, \sin 2t$, annihilated by $D(D^2 + 4)$) and $\text{Sym}^2(D^2 + D) = D^3 + 3D^2 + 2D$ (solutions $1, e^{-t}$; products $1, e^{-t}, e^{-2t}$, annihilated by $D(D+1)(D+2)$). These are `symSquare_golden_harmonic` and `symSquare_golden_exp` in `Agora/SymSquare.lean`; see §9.

5. IRREDUCIBILITY AND MINIMALITY

Everything in this section has status **(E)**: the arguments are exact over $\mathbb{Q}$, they involve no floating-point step, and they are reproduced by the checker `check_L3_irreducible_minimal.py`, which is shipped with four negative controls comprising sixteen assertions (§10). We treat s_7 throughout and record the s_{10} data in parallel, since the same argument applies verbatim.

We work over $\overline{\mathbb{Q}}(z)$; “irreducible” means that the operator admits no proper right factor of positive order in $\overline{\mathbb{Q}}(z)[D]$, equivalently that the associated differential module is simple. Since our operators are Fuchsian — all singular points regular — irreducibility of the differential module is equivalent to irreducibility of the monodromy representation [14, Ch. 5–6], and we pass between the two freely.

5.1. No hyperexponential solution. Recall that a nonzero function y is *hyperexponential* over $\overline{\mathbb{Q}}(z)$ if $y'/y \in \overline{\mathbb{Q}}(z)$. An order-2 operator is reducible precisely when it has a hyperexponential solution: a right factor of order 1 is $D - u$ with $u \in \overline{\mathbb{Q}}(z)$, and its solution $\exp \int u$ solves the order-2 operator; conversely a hyperexponential solution y produces the right factor $D - y'/y$.

Lemma 5.1 (Denominator obstruction; status (E)). *Let L be a Fuchsian operator of order 2 on $\mathbb{P}^1$ whose finite singular points and whose point $z = 0$ have all local exponents in $\frac{1}{2}\mathbb{Z}$, and suppose no exponent of L at ∞ lies in $\frac{1}{2}\mathbb{Z}$. Then L has no hyperexponential solution, and hence L is irreducible.*

Proof. Suppose y is a hyperexponential solution and put $u = y'/y \in \overline{\mathbb{Q}}(z)$. Because L is Fuchsian, y has moderate growth at every point of $\mathbb{P}^1$, so u has at worst simple poles and no polynomial part; write

$$u = \sum_p \frac{r_p}{z - p},$$

the sum over the finitely many poles. At a singular point p of L , the residue r_p is the local exponent of y at p , hence lies in $\frac{1}{2}\mathbb{Z}$ by hypothesis. At an ordinary point p of L , y is meromorphic near p and r_p is its vanishing order, a (possibly negative) integer; in fact solutions of a Fuchsian operator are holomorphic at ordinary points, so $r_p \in \mathbb{Z}_{\geq 0}$. Either way $r_p \in \frac{1}{2}\mathbb{Z}$ for every p , whence

$$\sum_p r_p \in \frac{1}{2}\mathbb{Z}.$$

The exponent of y at ∞ is $-\sum_p r_p$, and it must be one of the exponents of L at ∞ . But no exponent of L at ∞ lies in $\frac{1}{2}\mathbb{Z}$, a contradiction. Hence no hyperexponential solution exists, and by the remark above L is irreducible. $\square$

Proposition 5.2 (L_2 is irreducible; status (E)). *The s_7 partner L_2 of (5) is irreducible over $\overline{\mathbb{Q}}(z)$. The same holds for the s_{10} partner.*

Proof. By Proposition 3.2 the exponents of the s_7 partner are $\{0, 0\}$ at $z = 0$ and $\{0, \frac{1}{2}\}$ at each of $z = -1$ and $z = \frac{1}{27}$ — all in $\frac{1}{2}\mathbb{Z}$ — while at ∞ they are $\{\frac{1}{3}, \frac{2}{3}\}$, neither of which lies in $\frac{1}{2}\mathbb{Z}$. Lemma 5.1 applies. For s_{10} the finite data are identical in shape (exponents $\{0, \frac{1}{2}\}$ at $-\frac{1}{4}$ and $\frac{1}{16}$, $\{0, 0\}$ at 0) and the exponents at ∞ are $\{\frac{3}{8}, \frac{5}{8}\}$, again outside $\frac{1}{2}\mathbb{Z}$. $\square$

The checker computes the least common denominator N of all attainable residues at runtime rather than assuming it, obtaining $N = 2$ for both families, and then verifies that *every* exponent at ∞ is obstructed — not merely one of them.

5.2. The monodromy is not dihedral. Let V denote the rank-2 local system of solutions of L_2 on $\mathbb{P}^1 \setminus \{0, -1, \frac{1}{27}, \infty\}$, with monodromy group $G \subseteq \mathrm{GL}(V)$ and projective image $\overline{G} \subseteq \mathrm{PGL}_2$. We say the monodromy is *dihedral* if $\overline{G}$ is contained in the image of the normalizer $N(T)$ of a maximal torus $T \subset \mathrm{SL}_2$.

Lemma 5.3 (Nontrivial unipotent at the MUM point; status (E)). *The local monodromy of L_2 at $z = 0$ is unipotent and $\neq \mathrm{Id}$.*

Proof. The indicial equation of L_2 at $z = 0$ has the double root 0 (Proposition 3.2). By the Frobenius method a repeated indicial root forces a second solution containing $\log z$: with $A(z)$ the holomorphic solution normalized by $A(0) = 1$, a second solution is $A(z) \log z + g(z)$ with g holomorphic and $g(0) = 0$. Continuing once around $z = 0$ fixes A and adds $2\pi i A$ to the second solution, so the local monodromy is a single nontrivial Jordan block, i.e. unipotent and $\neq \text{Id}$. $\square$

Lemma 5.4 (Not dihedral; status (E)). *$\overline{G}$ is not contained in the image of the normalizer of a maximal torus of SL_2 .*

Proof. Let $g \in N(T)$. Either $g \in T$, in which case g is semisimple, or g lies in the nontrivial coset, in which case g conjugates T by inversion and is conjugate to an antidiagonal matrix; then $g^2 \in T$ is semisimple. Let u be a lift of the local monodromy at $z = 0$. By Lemma 5.3 u is unipotent and $\neq \text{Id}$, hence not semisimple; and u^2 is again unipotent and $\neq \text{Id}$, hence also not semisimple. So u lies in neither coset, and $\overline{G} \not\subseteq N(T)/\{\pm 1\}$. $\square$

5.3. Irreducibility of L_3 .

Lemma 5.5 (Symmetric square of a rank-2 system). *Let V be an irreducible rank-2 local system with monodromy group G and projective image $\overline{G}$. Then $\text{Sym}^2 V$ is reducible if and only if $\overline{G}$ is contained in the normalizer of a maximal torus.*

Proof. In characteristic 0, $V \otimes V \cong \text{Sym}^2 V \oplus \wedge^2 V$ and $V \otimes V \cong \text{End}(V) \otimes \det V$; comparing ranks and the trace decomposition $\text{End}(V) = \text{End}_0(V) \oplus \text{Id}$ gives

$$\text{Sym}^2 V \cong \text{End}_0(V) \otimes \det V,$$

so $\text{Sym}^2 V$ is reducible if and only if $\overline{G}$ acts reducibly on $\mathfrak{sl}(V) = \text{End}_0(V)$ by conjugation.

Suppose $\mathfrak{sl}(V)$ has a $\overline{G}$ -stable line, spanned by $X \neq 0$. If X is nilpotent, $\ker X \subset V$ is a line and is $\overline{G}$ -stable, contradicting irreducibility of V . Hence X is semisimple with two distinct eigenvalues, and $\overline{G}$ permutes its two eigenlines in V ; that is, $\overline{G}$ normalizes the maximal torus determined by that eigenline pair. If instead $\mathfrak{sl}(V)$ has a stable plane, its annihilator under the (nondegenerate, invariant) Killing form is a stable line and the previous case applies. The converse is immediate: if $\overline{G}$ normalizes T with eigenlines $\langle e_1 \rangle, \langle e_2 \rangle$, then the line spanned by $e_1 e_2$ in $\text{Sym}^2 V$ is stable. $\square$

Theorem 5.6 (L_3 is irreducible; status (E)). *The operator $L_3 = L_3(13, 4, -27, 3)$ is irreducible over $\overline{\mathbb{Q}}(z)$. The same holds for $L_3(6, 2, -64, 4)$.*

Proof. By Theorem 4.5, L_3 is $P_2 \cdot \text{Sym}^2(L_2/P_2)$, so its solution space is the space of products of solutions of L_2 , i.e. the local system $\text{Sym}^2 V$. V is irreducible by Proposition 5.2, and $\overline{G}$ is not contained in the normalizer of a maximal torus by Lemma 5.4. By Lemma 5.5, $\text{Sym}^2 V$ is irreducible, hence so is L_3 . $\square$

Corollary 5.7 (Minimality; status (E)). *L_3 is the minimal-order annihilator in $\overline{\mathbb{Q}}(z)[D]$ of each of its nonzero solutions; in particular of the holomorphic solution $u(z) = \sum_{n \geq 0} s_7(n)z^n$ at $z = 0$. Consequently the differential module generated by u has rank exactly 3.*

Proof. Let $y \neq 0$ solve $L_3 y = 0$ and suppose M has order m with $1 \leq m < 3$ and $My = 0$. Then $\gcd(L_3, M)$ annihilates y , so it has order ≥ 1 ; and it is a right factor of L_3 of order $\leq m < 3$. This is a proper right factor, contradicting Theorem 5.6. Hence no annihilator of order < 3 exists, and L_3 itself annihilates y . $\square$

Remark 5.8 (Why minimality is the load-bearing statement). Corollary 5.7 is what pins the rank of the differential module generated by the holomorphic solution at 3 rather than at 1 or 2. Every rank-based statement downstream — in particular the conditional rank statement of §8 — rests on it. Had L_3 factored, that rank, and everything computed from it, would move. This is why the argument is given in full here rather than cited.

Remark 5.9 (Negative controls; status (E)). The checker reproducing this section is deliberately fallible. Its controls feed it (i) an operator with a hyperexponential solution by construction, $\theta^2 - 2\theta + 1$, which annihilates $y = z$ so that $y'/y = 1/z$ is rational — the irreducibility step must and does reject it; (ii) an operator with distinct indicial roots at 0, so no logarithm and no unipotent — the non-dihedral step must and does reject it; (iii) a cross-check against the independently computed Riemann scheme of §3, requiring that the pairwise sums of the L_2 exponents reproduce the L_3 exponents at all four singular points for both families; and (iv) a positive control that the real operators still pass. All sixteen assertions behave as required.

6. THE MONODROMY LATTICE OF THE s_7 FAMILY

Disclosure. *The monodromy matrices used in this section are obtained numerically.* They are computed by analytic continuation at 60-digit working precision and then recognized as exact algebraic numbers. That recognition step is not a proof, and nothing derived from it is stated as a theorem here: the recognized matrices appear as Numerical Claim 6.2, status (N). Everything downstream of them — the invariant form, the orbit lattice, the Gram matrix, and the integral splitting — is exact integer and rational arithmetic, and is stated as a proposition *about the recognized matrices*, i.e. conditionally on Numerical Claim 6.2. The one statement in this section that is unconditional, because it is a finite verifiable assertion about an explicitly displayed integer matrix, is Proposition 6.4. We are equally explicit in §8 about what is *not* claimed: nothing here identifies the computed lattice with any invariant of a surface.

6.1. Numerical continuation. Work in the Frobenius basis of L_2 at the MUM point $z = 0$: the holomorphic solution $A(z)$ with $A(0) = 1$, and

$$\hat{B}(z) = \frac{A(z) \log z + g(z)}{2\pi i}, \quad g \text{ holomorphic, } g(0) = 0,$$

whose existence is Lemma 5.3. In this basis the local monodromy at $z = 0$ is $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ by construction.

The finite singular loci are computed at runtime as the roots of the leading θ -coefficient of L_2 , giving $\{-1, \frac{1}{27}\}$ for s_7 ; they are not typed in. Continuation is by Taylor stepping at 60 decimal digits of working precision, series order 140, step ratio 0.35, with the series tail bounded by 10^{-45} at every step and the bound asserted rather than assumed. Loops are taken based at $z_0 = \rho/4$, where ρ is the distance from 0 to the nearest singular point.

Remark 6.1 (Machinery control). Before any unknown loop is computed, the numerical machinery is required to reproduce a matrix known analytically: the continued loop around $z = 0$ must return $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. It does, to 3.17×10^{-61} . A pipeline that cannot recover a known answer is not used to produce unknown ones.

Numerical Claim 6.2 (Elliptic monodromies of the s_7 partner; status (N)). In the Frobenius basis above, the monodromy matrices of L_2 around the two finite singular points are

$$M(\frac{1}{27}) = \frac{i}{\sqrt{7}} \begin{pmatrix} 0 & 1 \\ -7 & 0 \end{pmatrix}, \quad M(-1) = \frac{i}{\sqrt{7}} \begin{pmatrix} 7 & 4 \\ -14 & -7 \end{pmatrix}.$$

The evidence, and the limits of that evidence, are as follows. The two numerically continued matrices satisfy $\det M = -1$ and $M^2 = \text{Id}$ to about 10^{-59} , as the local exponents $\{0, \frac{1}{2}\}$ of Proposition 3.2 require. The quantity $\sqrt{7}$ is not put in by hand: it emerges from the continuation. What is actually recognized is not the 2×2 matrices themselves but their symmetric squares in the flag basis $(A^2, A\hat{B}, \hat{B}^2)$, which are rational; recognition is performed with a loud acceptance gate of 10^{-35} , the largest observed residual is $\approx 10^{-59}$, and the denominators appearing in the recognized entries come out as $\{1, 7\}$. The recognized matrices are then subjected to exact post-verification: each is an involution; the product of the three loops matches the monodromy at ∞ predicted by the independently computed exponents $\{\frac{1}{3}, \frac{2}{3}\}$ there (trace 0, order 3); the joint invariant symmetric bilinear form exists and is unique up to scale; and the orbit lattice below closes under all generators and their inverses. A conspiracy of numerical errors surviving all of these exact gates is not credible. It is nevertheless not a proof, which is why this is a Numerical Claim.

6.2. The exact lattice pipeline. From here on, all arithmetic is exact — integers and rationals only.

Proposition 6.3 (The joint monodromy-invariant lattice; status (E), given Numerical Claim 6.2). *Let $N(p) = \text{Sym}^2 M(p)$ denote the recognized rational*

3×3 matrices and let Γ be the group they generate together with the cusp matrix $N(0) = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$. Then:

- (1) the space of Γ -invariant symmetric bilinear forms on $\mathbb{Q}^3$ is one-dimensional;
- (2) the Γ -orbit of the cusp-invariant isotropic vector f spans a lattice that stabilizes after finitely many steps and is closed under all generators and inverses;
- (3) in its primitive even rescaling, the restriction of the invariant form to that lattice has Gram matrix

$$G = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 14 & 0 \\ -1 & 0 & 0 \end{pmatrix},$$

with $\det G = -14$, signature $(2, 1)$, discriminant group $\mathbb{Z}/14$ and discriminant form (in the sense of [9]) taking the value $q = \frac{1}{14} \bmod 2\mathbb{Z}$ on a generator;

- (4) the divisibility of $(N(0) - \text{Id})^2$ on the lattice yields $2n = 14$;
- (5) the lattice admits no proper even Γ -invariant overlattice.

Items (i)–(v) are finite exact computations on the recognized matrices; they are performed by `check_U1_lattice.py`, whose structural conditions are asserted (and therefore can fail) rather than reported. The sign of the form is fixed by an independent positivity re-derivation of the framework period shape, not chosen.

6.3. The integral splitting. The next statement is the one place in this section where nothing is conditional. It is a finite assertion about the explicit integer matrix G displayed above, checkable by hand in a minute, and we give the witness so that the reader can do so.

Proposition 6.4 ($G \cong U \oplus \langle 14 \rangle$ over $\mathbb{Z}$; status (E), unconditional). *Let G be the integer matrix of Proposition 6.3(iii) and let*

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}, \quad \det P = -1.$$

Then $P \in \text{GL}_3(\mathbb{Z})$ and

$$P^\top G P = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 14 \end{pmatrix} = U \oplus \langle 14 \rangle.$$

In particular the lattice with Gram matrix G is isometric over $\mathbb{Z}$ to $U \oplus \langle 14 \rangle$.

Proof. Write p_1, p_2, p_3 for the columns of P and $\langle \cdot, \cdot \rangle$ for the form with Gram matrix G in the standard basis e_1, e_2, e_3 . Then $p_1 = e_1$, $p_2 = -e_3$, $p_3 = -e_2$, and directly from G : $\langle p_1, p_1 \rangle = G_{11} = 0$; $\langle p_1, p_2 \rangle = -G_{13} = 1$; $\langle p_1, p_3 \rangle = -G_{12} = 0$; $\langle p_2, p_2 \rangle = G_{33} = 0$; $\langle p_2, p_3 \rangle = G_{32} = 0$; $\langle p_3, p_3 \rangle = G_{22} = 14$. So

$\{p_1, p_2\}$ spans a hyperbolic plane U , orthogonal to $\langle p_3 \rangle \cong \langle 14 \rangle$. Expanding $\det P$ along the first column gives $\det \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} = -1$, so P is unimodular and the base change is an isometry of lattices over $\mathbb{Z}$. $\square$

Remark 6.5 (Two independent verifications). Proposition 6.4 has been verified by two implementations written independently of one another and sharing no code.

- (1) The pipeline of `check_U1_lattice.py` constructs a base change of determinant $+1$ during its stage 3 and checks $S^\top GS = U \oplus \langle 14 \rangle$ exactly. It serializes $\det S$ and $S^\top GS$ to its certificate but not S itself.
- (2) A from-scratch implementation, `check_U1_splitting_independent.py`, loads the certificate at runtime, reads only G and the integer d (never hardcoding 14 outside a labeled control), assembles the target $U \oplus \langle d \rangle$ at runtime, and re-derives its own witness from G alone — locating a primitive isotropic vector of unimodular pairing, completing it to a hyperbolic pair by an extended-gcd combination, and taking the orthogonal-complement generator. All arithmetic is exact Python integers. It returns the matrix P displayed in Proposition 6.4, of determinant -1 .

The two witnesses differ (determinants $+1$ and -1); that they differ is expected, since any element of $\mathrm{GL}_3(\mathbb{Z})$ realizing the isometry is an equally valid proof, and it confirms that the second implementation did not reproduce the first. The second is shipped with five negative controls: perturbing one entry of G must break agreement with the certificate’s own determinant, and does; corrupting the witness must break $P^\top GP = U \oplus \langle d \rangle$, and does; doubling a column of the witness must be rejected at the $\mathrm{GL}_3(\mathbb{Z})$ gate, and is; testing the genuine s_7 Gram against the target $U \oplus \langle 20 \rangle$ must fail, and does; and the unmodified certificate must still pass after all scrambling hooks have been exercised, and does.

6.4. The s_{10} control.

Numerical Claim 6.6 (Level discrimination; status (N)+(E)). The identical pipeline — same code, same tolerances, same exact post-verification — applied to the s_{10} family derives determinant -20 , $2n = 20$, and an integral splitting $U \oplus \langle 20 \rangle$.

This is the control that matters most. A pipeline that produced $U \oplus \langle 14 \rangle$ from any input, or that had 14 wired into it anywhere, would be worthless; running it unchanged on a different member of the same template and obtaining a different, internally consistent answer at a different level shows that the computation is sensitive to the operator it is given. The certificate records the s_{10} values as computed-versus-computed inequalities against the s_7 values, with no expected result typed in. Further controls in the same suite corrupt the recognized matrices in three distinct ways: a raw entry

perturbation fails at the involution gate; a conjugated involution fails at the infinity-product gate; and a corruption fed directly to the invariant-form step causes the joint invariant form to vanish (solution space of dimension 0) rather than to be silently replaced by a different form.

Remark 6.7 (What has and has not been established). Established, conditionally on Numerical Claim 6.2: the joint monodromy-invariant lattice of the s_7 family, in its primitive even scaling, is isometric over $\mathbb{Z}$ to $U \oplus \langle 14 \rangle$, with the discriminant data of Proposition 6.3. Established unconditionally: Proposition 6.4, a statement about the displayed matrix G . *Not* established anywhere in this section: that this lattice is the transcendental lattice of anything. That identification is a separate question, taken up — and left conditional — in §8.

7. THE ARITHMETIC SEPARATION OF THE THREE CANDIDATES

Theorem 4.5 is blind to the choice of parameters. This section exhibits the invariant that is not: the integrality of the holomorphic solution of the order-2 partner. Among Cooper's three sporadic sequences, only s_7 has an integral partner, and we make the mechanism behind that integrality precise.

7.1. The partner recurrence. Extracting the coefficient of z^M from $L_2 f = 0$ for $f = \sum_{n \geq 0} a_n z^n$ with $a_0 = 1$, using the partner coefficients (9), gives $a_1 = \frac{b}{2}$ and, for $k \geq 0$,

$$(13) \quad (k+2)^2 a_{k+2} = a_{k+1} \left(2a(k+1)^2 + a(k+1) + \frac{b}{2} \right) - a_k \left(ck^2 + ck + \frac{c+d}{4} \right).$$

The leading coefficient $(k+2)^2$ is the crux: the sequence is $\mathbb{Q}$ -valued by construction, and integrality — when it holds — is an Apéry-style theorem, not a consequence of the shape of the recursion.

There is, however, a floor. Let $s: \mathbb{N} \rightarrow \mathbb{Z}$ be *any* integer sequence with $s(0) = 1$ and let g be the coefficient sequence of the formal square root of $\sum_n s(n)z^n$, defined by $g_0 = 1$ and $g_n = (s(n) - \sum_{k=1}^{n-1} g_k g_{n-k})/2$.

Proposition 7.1 (Dyadic baseline; status (K)). *For every integer sequence s with $s(0) = 1$, every g_n lies in $\mathbb{Z}[\frac{1}{2}]$.*

Proof. Induction on n : $\mathbb{Z}[\frac{1}{2}]$ is a ring containing $\mathbb{Z}$, so the sum of products of earlier terms lies in it, and it is closed under division by 2. Formalized as `sqrSeq_dyadic` in `Agora/Sequences/FormalSqrt.lean`; the accompanying control `dyadic_baseline_control` checks that $\frac{1}{3} \notin \mathbb{Z}[\frac{1}{2}]$, so the statement is not vacuous. $\square$

So powers of 2 in the denominators are the *forced* baseline for the partner of any Cooper-template instance with integral holomorphic solution, and s_{10} and s_{18} merely exhibit it. It is s_7 that is anomalous.

7.2. Non-integrality for s_{10} and s_{18} .

Theorem 7.2 (status (K)). *The holomorphic solution of the s_{10} partner has coefficients beginning $1, 1, \frac{17}{2}, \frac{147}{2}$, and that of the s_{18} partner begins $1, 3, \frac{45}{2}, \frac{429}{2}$. In particular neither sequence is integral.*

Proof. Evaluate (13) at the respective parameters; the terms listed are exact rationals. Integrality of the $n = 2$ term would force $17 = 2m$ resp. $45 = 2m$ for an integer m , which is impossible. The formalization states the refutation in this arithmetic form rather than as denominator bookkeeping: `s10_partner_two_not_integral`, `s18_partner_two_not_integral`, and the conclusions `s10_partner_not_integral`, `s18_partner_not_integral` in `Agora/Sequences/PartnerIntegrality.lean`. $\square$

Note the asymmetry of difficulty: refuting integrality needs one witness, establishing it needs all of them. The formalization keeps that asymmetry visible in the type of its summary statement (`cooper_partner_separation`), which asserts only the two negative halves.

7.3. Integrality for s_7 . For $(a, b, c, d) = (13, 4, -27, 3)$, recurrence (13) becomes, after clearing,

$$(14) \quad (k+2)^2 a_{k+2} = (26(k+1)^2 + 13(k+1) + 2) a_{k+1} + 3(3k+1)(3k+2) a_k, \quad a_0 = 1, a_1 = 2.$$

Its first terms are

$$1, 2, 22, 336, 6006, 117348, 2428272, 52303680, \dots$$

which is OEIS A279619 [11]. Integrality of this sequence is a theorem of the literature, and we state it as such.

Theorem 7.3 (cited, O’Brien 2016). [10, Thm. 6.2, p. 47]. *Any rational sequence satisfying recurrence (14) with the stated initial values is integer-valued.*

O’Brien’s proof is an explicit coefficient-matching induction using the q -expansions of two specific weight-1 level-7 modular objects; it is not a formal consequence of any general integrality principle, and §7.4 explains why. We do not reprove it.

Corollary 7.4 (status (K), modulo Theorem 7.3). *The holomorphic solution of the s_7 partner is integral, and its coefficient sequence is A279619.*

Proof. That the sequence defined by the generic recurrence (13) at the s_7 parameters satisfies exactly (14), with the same initial values, is a mechanical identity; it is kernel-checked as `partnerSeq_s7_recurrence`. Theorem 7.3 then applies. In the formalization the cited theorem is the registered axiom `obrien2016_theorem6_2`, and the conclusion is `s7_partner_integral`; see §9 for the axiom’s statement and its disclosure. $\square$

Remark 7.5 (PASS(7)). The formalization independently retains a kernel-checked finite verification, `s7_partner_integral_pass7`: the coefficients are integers for $n \leq 7$, each obtained by evaluating the recurrence rather than by table lookup, with the eight witnesses displayed above. This was the honest position before the citation was traced to its correct source, and it is kept on the record. It is evidence, not proof, of the general statement, and is reported as *PASS(7)* rather than as a theorem about all n (§9).

7.4. The mechanism: a normalized integral uniformizer. Why is s_7 's partner integral when the dyadic floor of Proposition 7.1 permits denominators? The answer is not that η -quotients have integer Fourier coefficients. It is a normalization statement, and the distinction is not pedantic — the object in question is a series *reversion*, and reversion does not preserve integrality in general.

Following [10], set

$$(15) \quad z_7 = \sum_{m,n \in \mathbb{Z}} q^{m^2+mn+2n^2}, \quad Z_7 = \frac{7P(q^7) - P(q)}{6}, \quad X_7 = \frac{\eta_1^3 \eta_7^3}{z_7^3},$$

with P the Ramanujan Eisenstein series $P(q) = 1 - 24 \sum_{n \geq 1} \sigma(n)q^n$ and $\eta_1^3 \eta_7^3 = q \prod_{j \geq 1} (1 - q^j)^3 (1 - q^{7j})^3$. O'Brien's Theorem 6.1 gives the expansion $z_7 = \sum_{n \geq 0} c_7(n) X_7^n$, and the $c_7(n)$ are the terms of (14).

Proposition 7.6 (Reversion of a normalized integral series). *Let $X \in q\mathbb{Z}[[q]]$ have leading coefficient exactly 1, i.e. $X = q + O(q^2)$, and let $Y \in \mathbb{Z}[[q]]$. Then the unique expansion $Y = \sum_{n \geq 0} c_n X^n$ has $c_n \in \mathbb{Z}$ for all n .*

Proof. Determine the c_n recursively: $c_0 = Y(0)$, and having fixed $c_0, \dots, c_{n-1}$, the residual $Y - \sum_{k < n} c_k X^k$ lies in $q^n \mathbb{Z}[[q]]$, while $X^n = q^n + O(q^{n+1})$; so c_n is the coefficient of q^n in the residual divided by 1, hence an integer. Integrality of the leading coefficient of X is exactly what makes that division trivial; a leading coefficient of 2, say, would reintroduce denominators of precisely the kind Proposition 7.1 permits. $\square$

Proposition 7.7 (The mechanism, verified; status (E)). *With X_7 as in (15), exact rational arithmetic on q -expansions gives*

$$X_7 = q - 9q^2 + 30q^3 - 15q^4 + \dots \in q\mathbb{Z}[[q]],$$

so X_7 is a normalized integral uniformizer; and expanding z_7 in powers of X_7 reproduces 1, 2, 22, 336, 6006, 117348, 2428272, i.e. A279619. Consequently, by Proposition 7.6, integrality of c_7 follows from (15) together with O'Brien's Theorem 6.1 — but from those taken jointly, and not from the integrality of an η -quotient alone.

Remark 7.8 (Resolving the source's equation (3.15); status (E)). The pinned copy of [10] renders the denominator of its equation (3.15) ambiguously between z_7 and Z_7 and drops the exponent, so which object is meant is not

determined by the text. Three readings were tested in exact arithmetic:

$$\frac{\eta_1^3 \eta_7^3}{z_7^3}, \quad \frac{\eta_1^3 \eta_7^3}{Z_7^3}, \quad \frac{\eta_1^3 \eta_7^3}{Z_7^2}.$$

All three are normalized integral uniformizers, and all three therefore produce integral expansion coefficients — which is exactly why integrality alone cannot resolve the ambiguity. They produce, respectively,

$$1, 2, 22, 336, 6006, 117348, 2428272; \quad 1, 2, 34, 828, 23662, 739984, 24521448; \quad 1, 2, 26, 476, 10206, 23$$

so only the first reproduces A279619. It is also the only weight-consistent reading: $\eta_1^3 \eta_7^3$ has weight 3 and z_7 has weight 1, so z_7^3 is what makes the quotient weight 0, as a uniformizer must be. The computation additionally verifies $z_7^2 = Z_7$ exactly, confirming that the two symbols denote genuinely different objects. The resolution is recorded so that the ambiguity need not be re-litigated.

Remark 7.9 (Scope). Proposition 7.7 establishes nothing about s_{10} or s_{18} . Their partners are non-integral by Theorem 7.2, and no modular parametrization by a normalized integral uniformizer is known for them — but absence of a known parametrization is not a proof that none exists, and no such negative claim is made here. In particular no level is assigned to s_{18} anywhere in this paper.

7.5. The modular coordinate. The parameter of the s_7 family is a modular function for the same group that Remark 3.3 identified from the exponent data.

Proposition 7.10 (status (E), with a cited group-theoretic input). *Let t be the generating function of OEIS A279618 [11] and let $u = q^{-1}(E(q)/E(q^7))^4$, with $E(q) = \prod_{n \geq 1} (1 - q^n)$, be the classical $\Gamma_0(7)$ Hauptmodul; put $v = 1/u \in q\mathbb{Z}[[q]]$. Then, in exact rational arithmetic on q -expansions:*

- (1) t is not a Möbius function of v (the linear system solved from the first 6 orders admits no solution verifying on later orders);
- (2) t is of the form $(\alpha v^2 + \beta v + \gamma)/(\delta v^2 + \epsilon v + \zeta)$, with coefficients solved linearly from the first 12 orders and then verified on held-out orders 13 and 29 that were not used in solving (29 orders available in total);
- (3) rewriting t as a degree-2 rational function of u and solving for the involution constant κ with $R(\kappa/u) \equiv R(u)$ yields $\kappa = 49$.

Items (i) and (ii) give $[\mathbb{Q}(u) : \mathbb{Q}(t)] = 2$, so $\mathbb{Q}(t)$ is the fixed field of an element normalizing $\Gamma_0(7)$. The normalizer of $\Gamma_0(N)$ in $\mathrm{PSL}_2(\mathbb{R})$ for prime N is the Fricke extension $\Gamma_0(N) + [2]$ — this step is cited, not re-derived — so that element is the Fricke involution and t is a $\Gamma_0(7) +$ Hauptmodul. Item (iii) exhibits the Fricke constant $\kappa = 7^2$ directly.

The constant κ is computed from the fitted coefficients, never assumed; the checker’s verdict requires it to come out 49 but does not supply that value to the fit. Three negative controls accompany the computation: feeding it v itself

(a genuine $\Gamma_0(7)$ Hauptmodul) must return the verdict “not plus”, and does; corrupting one coefficient of A279618 must break the degree-2 verification, and does; and testing against the wrong level, $v_5 = q(E(q^5)/E(q))^6$ for $\Gamma_0(5)$, must fail every fit, and does.

Remark 7.11 (Three independent indicators of the level). The number 7 has now appeared from three computations that share no code: the implied Fuchsian signature of the s_7 operators, matching $\Gamma_0(7)+$ (Remark 3.3, status (E)); the Hauptmodul and Fricke constant $\kappa = 49$ of Proposition 7.10 (status (E)); and the $\sqrt{7}$ appearing uninvited in the numerically continued monodromy, together with $\det G = -14$ (Numerical Claim 6.2 and Proposition 6.3, status (N)/(E)). What that convergence does and does not license is the subject of §8.

8. RELATION TO THE M_n -POLARIZED FRAMEWORK

This section places the computations of §§4–7 next to a classical framework and states carefully what the juxtaposition does and does not establish. Nothing in this section is claimed as a new theorem.

8.1. The framework, as cited.

Theorem 8.1 (cited, Dolgachev 1996). [5, §7, p. 20 and Thm. 7.1]. *For $M_n = U \oplus E_8^2 \oplus \langle -2n \rangle$ one has $(M_n)^\perp = U \oplus \langle 2n \rangle$ in the K3 lattice, and the coarse moduli space of M_n -polarized K3 surfaces is the Fricke modular curve $\mathbb{H}/\Gamma_0(n)+$.*

Theorem 8.2 (cited, Doran 2000). [6, Thm. 5.13]. *The Picard–Fuchs operator of an M_n -polarized family of K3 surfaces is the symmetric square of a second-order Fuchsian operator.*

Both sources were fetched, hash-pinned, and read by the program; the parts of Theorem 8.1 used below were additionally re-derived symbolically before use, with the period shape and the invariant bilinear form recomputed from scratch rather than transcribed. We also record, because it governs how much the coincidences below can be made to prove, that Doran himself notes the classification of rank-19 lattice polarizations to be incomplete [6, §6].

8.2. The logical direction. A referee’s first reaction to §6 should be that Theorem 8.1 already says $(M_7)^\perp = U \oplus \langle 14 \rangle$ and Theorem 8.2 already says the Picard–Fuchs operator of such a family is a symmetric square, so nothing has been learned. The reply is that the direction of inference here is the reverse of the framework’s. The framework starts from a polarization hypothesis and deduces invariants. We start from a concretely given operator family — specified by four rational numbers — and *compute* its invariants: the symmetric-square structure (kernel-checked, and for the whole template rather than for one member), the Fuchsian signature, the modular coordinate

and its level, and the monodromy lattice. Each of these is a fact about the operator, established without any geometric hypothesis. The framework then supplies a reading under which they all cohere, and the coherence is close: the same level 7 appears from four computations that share no code, and the monodromy lattice matches $(M_7)^\perp$ on the nose, including the discriminant form.

That is convergent evidence, not a proof, and the incompleteness of the rank-19 classification is precisely why it cannot be upgraded by citation.

Computed fact about the s_7 operators	Status	Framework counterpart
$L_3 = P_2 \cdot \text{Sym}^2(L_2)$, uniformly on the template (Thm. 4.5)	(K)	Thm. 8.2
Almkvist–van Straten $W \equiv 0$ (Thm. 4.9)	(K)	self-adjointness of a Sym^2
L_2, L_3 irreducible; L_3 minimal (§5)	(E)	rank-3 sub-system
Implied Fuchsian signature: genus 0, orders $(2, 2, 3)$, one cusp, $\text{area}/2\pi = \frac{2}{3}$ (Rem. 3.3)	(E)	signature of $\Gamma_0(7)+$
Modular coordinate is a $\Gamma_0(7)+$ Hauptmodul, $\kappa = 49$ (Prop. 7.10)	(E)	$\mathbb{H}/\Gamma_0(7)+$ of Thm. 8.1
$\sqrt{7}$ in the elliptic monodromies (NC 6.2)	(N)	level 7
Monodromy lattice $U \oplus \langle 14 \rangle$, $\det = -14$, disc. $\mathbb{Z}/14$, $q = \frac{1}{14}$ (Props. 6.3, 6.4)	(N)+(E)	$(M_7)^\perp$ of Thm. 8.1
s_{10} control gives $U \oplus \langle 20 \rangle$ (NC 6.6)	(N)+(E)	level 10, not level 7

TABLE 1. Computed invariants of the s_7 operator family against the M_n -polarized framework. Status labels as in §1.

8.3. The evidence, assembled.

8.4. A conditional rank statement. We now record the rank statement that the minimality result of §5 yields *if* two external inputs are granted. It is stated as a conditional proposition, with its hypotheses displayed, and it is used nowhere else in this paper.

Proposition 8.3 (conditional; status (C)+(E) under (H1)–(H2)). *Assume:*
(H1) *the s_7 family is realized by the explicit projective K3 model of Almkvist and van Straten [1], and L_3 is the Picard–Fuchs operator of the period of its holomorphic 2-form; and*

(H2) *for the very general member X of the family, $\text{rank } T(X)$ equals the rank of the $\mathbb{Q}$ -sub-variation of Hodge structure generated by that period under Gauss–Manin.*

Then $\text{rank } T(X) = 3$ and $\rho(X) = 19$.

Proof. By Corollary 5.7, the sub-variation generated by the period has rank 3, that being the order of its minimal annihilator. (H2) transfers this to $\text{rank } T(X)$, and $\rho(X) = 22 - \text{rank } T(X) = 19$. $\square$

Remark 8.4 (On (H2), and on the caveat that must travel with $\rho = 19$). (H2) is standard Hodge theory and we cite rather than re-derive it. It is the conjunction of two inclusions. First, for very general z the Néron–Severi lattice is locally constant, the Noether–Lefschetz locus being a countable union of proper subvarieties, so $T = \text{NS}^\perp$ is a Gauss–Manin-flat sub-Hodge-structure containing the holomorphic 2-form; the sub-variation generated by that form is therefore contained in T . Second, $T(X) \otimes \mathbb{Q}$ is by definition the minimal primitive sub-Hodge-structure of $H^2(X, \mathbb{Q})$ whose complexification contains $H^{2,0}$, and the saturation of the sub-variation is such a structure; so the reverse inclusion holds [8, ch. 3 §3.1, ch. 17 §17.1]. Simplicity of the transcendental Hodge structure is [15, Thm. 1.6(a)].

The caveat is essential and is not a formality: $\rho = 19$ is the value for the *very general* member. It jumps to 20 on a countable dense subset of the base. Any use of Proposition 8.3 that drops the phrase “very general” is a misuse of it.

8.5. What is conjectural.

Conjecture 8.5 (Identification of the computed lattice). *The joint monodromy-invariant lattice computed in §6 is the transcendental lattice of the very general member of the s_7 family; that is, $T \cong U \oplus \langle 14 \rangle$, and the family is M_7 -polarized.*

We are explicit about the two gaps between Proposition 6.3 and Conjecture 8.5, because within the computation itself the identification is pinned only up to two enumerable ambiguities, and only one of them is closed by computation.

- (1) *Even invariant overlattices.* A priori the transcendental lattice could be an even overlattice of the computed orbit lattice, with the same rational form. These were enumerated: there are none (Proposition 6.3(v)). This gap is closed, conditionally on Numerical Claim 6.2.
- (2) *Overall integral rescaling.* The transcendental lattice could carry $\lambda \cdot G$ for an integer $\lambda > 1$. This branch is excluded by the shape $U \oplus \langle 2n \rangle$ that the framework predicts (Theorem 8.1) — that is, by a reading, not by a computation. It is the honest residual gap, and we do not paper over it: the argument that rules it out presupposes the conclusion it is used to support.

To these must be added the fact that Numerical Claim 6.2 is itself numerics-backed, and hypothesis (H1) of Proposition 8.3, which is a citation about a geometric model and is not verified anywhere in this paper.

8.6. What is not claimed. We close by listing, explicitly, statements a reader might expect to find here and will not.

- No degeneration or fibre-type reading is made of the singular points $z = -1$ and $z = \frac{1}{27}$. Remark 3.3 identifies them as order-2 elliptic points of the quotient curve; that is the only reading taken, and no elliptic fibration is invoked anywhere.
- No claim is made that the s_{10} or s_{18} partners are modular, and no level is assigned to s_{18} (Remark 7.9).
- Theorem 4.5 is not claimed to imply Theorem 8.2 or to be implied by it: the template theorem exhibits a partner from four rational parameters with no geometric hypothesis, whereas Doran’s theorem asserts existence under a polarization hypothesis. Neither subsumes the other.
- Proposition 8.3 is not asserted unconditionally, and Conjecture 8.5 is not asserted at all.

8.7. Open questions.

- (1) Make Numerical Claim 6.2 exact. The natural route is exact hypergeometric connection formulae for L_2 , which would upgrade Propositions 6.3 and 6.4 to unconditional statements about the s_7 family and remove the only numerical input in this paper.
- (2) Close the rescaling ambiguity of Conjecture 8.5 by a computation rather than by the framework’s shape — for instance by exhibiting a $\lambda > 1$ obstruction intrinsic to the monodromy representation.
- (3) Prove Theorem 7.3 in a proof assistant, discharging the one literature axiom of the formal development (§9). This requires q -expansion machinery for weight-1 level-7 modular objects that the pinned library does not have.
- (4) Determine whether the s_{18} partner exhibited in Corollary 4.7 — which appears not to have been written down before — has any modular interpretation, or whether its non-integrality is an obstruction to one.
- (5) Establish the genus statement for the abstract lattice $U \oplus \langle 14 \rangle$ independently. It is not needed for Proposition 6.4, which is constructive and therefore stronger than a genus argument for the specific lattice at hand, but it would be the natural framing for a classification statement.

9. THE FORMAL DEVELOPMENT

The statements labeled **(K)** in this paper are theorems of a Lean 4 development which we describe here in enough detail to be audited. We

also describe what the kernel did *not* check, since a formalization paper that reports only its successes is of no use to a referee.

9.1. Toolchain and layout. The development builds with Lean 4 at toolchain v4.32.0 [4] against `mathlib4` [13, 12] pinned at commit 3dffaf2f18b47d11948f6390838ea6f2. It is organized so that each epistemic category has exactly one place to live:

Directory	Contents and rule
<code>Agora/</code>	The mathematics. No <code>sorry</code> anywhere.
<code>Agora/Axioms/</code>	The only permitted location for an <code>axiom</code> declaration.
<code>OpenGoals/</code>	The only permitted location for a <code>sorry</code> ; each is a named goal.
<code>Tests/</code>	Golden numeric tests against independently computed values.

The placement rules are enforced mechanically by a repository hook rather than by convention: an `axiom` introduced outside `Agora/Axioms/` is rejected, and the no-`sorry` check excludes `OpenGoals/` and nothing else. This matters because the failure mode being guarded against is not dishonesty but drift — an assumption introduced locally during a proof attempt and never surfaced.

9.2. The results, and how they are stated. The kernel-checked content of this paper is:

Statement	Lean name	File
Thm. 4.5, θ^3	<code>partner_res3</code>	<code>Sequences/PartnerOperators.lean</code>
Thm. 4.5, θ^2	<code>partner_res2</code>	"
Thm. 4.5, θ^1	<code>partner_res1</code>	"
Thm. 4.5, θ^0	<code>partner_res0</code>	"
Lemma 4.4	<code>partner_magic</code>	"
Lemma 4.3	<code>cooperC2_eq_thetaC_cooperC3</code>	"
Cor. 4.7	<code>s7_P2_eq, ..., s18_P0_eq</code>	"
Thm. 4.9	<code>P_cleared_eq_zero</code>	<code>Swampland/SymSquareC3b.lean</code>
Rem. 4.12	<code>symSquare_golden_harmonic/_exp</code>	<code>SymSquare.lean</code>
Prop. 7.1	<code>sqrtSeq_dyadic</code>	<code>Sequences/FormalSqrt.lean</code>
Thm. 7.2	<code>s10_partner_not_integral, s18_...</code>	<code>Sequences/PartnerIntegrality</code>
Cor. 7.4, mechanical half	<code>partnerSeq_s7_recurrence</code>	"
Cor. 7.4	<code>s7_partner_integral</code>	"
Rem. 7.5	<code>s7_partner_integral_pass7</code>	"

Three design decisions shape how these are stated.

θ -form, not monic D -form. Normalizing an order-3 operator to be monic in D introduces genuine rational-function coefficients, and the rational-function type in the pinned library carries no derivative API. Rather than improvise one, every statement above lives in `Polynomial  $\mathbb{Q}$` : no division, no nonvanishing side conditions, and every proof obligation a polynomial identity. This is why Theorem 4.5 is stated as four coefficient identities plus the collapse lemma rather than as an operator equation — the factorization

in $\mathbb{Q}(z)[\theta]$ is then derived on paper, as in §4, from kernel-checked polynomial facts.

Concrete identities, not existence claims. The relation “ L_3 is the symmetric square of L_2 ” is encoded as an equality fixing *every* coefficient of L_3 as an explicit polynomial function of those of L_2 . It is falsifiable per candidate. The development had an earlier generation of statements of the form “there exists a polynomial of degree 3”, true of any cubic and therefore content-free; those were deleted and the episode is recorded in the repository’s escalation log. The lesson generalizes: a formalized statement is worth exactly the content of its statement, and no amount of kernel checking supplies content the statement lacks.

Derivatives are computed, not transcribed. Every occurrence of a derivative in Theorem 4.9 is an application of the library’s `Polynomial.derivative`, so the kernel performs each differentiation itself. Encoding a hand-computed derivative as an opaque definition would let a transcription error pass `ring` undetected.

9.3. The axiom inventory, in full. The development registers exactly one load-bearing literature axiom.

```
axiom obrien2016_theorem6_2 (c : N -> Q) (h0 : c 0 = 1) (h1 : c 1 = 2)
  (hrec : forall k : N,
    ((k : Q) + 2)^2 * c (k + 2) =
      (26 * ((k : Q) + 1)^2 + 13 * ((k : Q) + 1) + 2) * c (k + 1)
      + 3 * (3*(k : Q) + 1) * (3*(k : Q) + 2) * c k) :
  forall n, exists m : Z, c n = (m : Q)
```

This is Theorem 7.3 verbatim, re-indexed by $n = k + 1$ to avoid truncated subtraction on $\mathbb{N}$ and with the source’s boundary computation replaced by the two initial values as hypotheses. It is used exactly once, to conclude `s7_partner_integral` from `partnerSeq_s7_recurrence`. It is registered in the repository’s axiom register with its provenance, and the source was read in full before being cited — which mattered: an earlier attribution in the program named the wrong theorem of the same source and a mechanism (“ η -quotients have integer coefficients, hence a corollary”) that does not in fact establish integrality, since the object is a series reversion. See Proposition 7.6 for what the correct mechanism is.

It is not provable in the pinned library: O’Brien’s argument needs the q -expansions of two specific weight-1 level-7 modular objects, obtained by classical modular-forms machinery that `mathlib` at the pinned commit does not contain. Discharging it is open question 3 of §8.

For completeness: the register carries one further axiom declaration, `pipeline_upper_bound`, belonging to a part of the repository unrelated to the mathematics of this paper. It is disclosed in the register as vacuously true — its statement is an existential satisfied by the witness 1 and encodes no data — and no file used in this paper imports it. We mention it because an axiom inventory that reports only the axioms one wishes to discuss is not an inventory.

9.4. Golden validation and $\text{PASS}(N)$ discipline. Two devices keep the development from certifying its own mistakes.

Golden tests check a formula against an answer obtained by other means. The symmetric-square formula (3) is validated against two examples whose answers are derivable by inspection (Remark 4.12); no candidate-level symmetric-square theorem is accepted until those are green, because the formula is the single point of failure for every such theorem. The sequence values in `Tests/` are checked against reference values computed independently in exact integer arithmetic outside Lean, for $n \leq 19$. Separately, the recurrence-defined partner sequence is checked against the formal square root of the bulk sequence, on both the integral side (s_7 : 2, 22, 336) and the non-integral side (s_{10} : $1, \frac{17}{2}$) — so the $\frac{17}{2}$ witnessing Theorem 7.2 is not an artifact of one encoding.

$\text{PASS}(N)$ is the reporting convention for a kernel-checked finite range. `s7_partner_integral_pass7` verifies integrality for $n \leq 7$ and is cited as $\text{PASS}(7)$, never as “the sequence is integral” — a distinction that is not pedantry here, since establishing integrality needs every term while refuting it needs one. The convention survived the arrival of the axiom-backed general theorem: the finite statement was kept rather than deleted, and the summary theorem `cooper_partner_separation` deliberately asserts only the two *negative* halves of the separation, so that no downstream file can cite “only s_7 is integral” as a single uniformly-established fact.

9.5. What the kernel did not check.

- *Theorem 7.3*, as discussed. Kernel-checked: that our sequence satisfies its hypotheses.
- *Everything with status (E) or (N)*: §5, §6, and the computational parts of §7 are Python and computer-algebra computations, not Lean. They are exact and reproducible (§10) but they are not kernel-checked, and the paper never labels them as though they were.
- *The golden tests in Tests/* use `native_decide` for terms up to $n = 19$, whose binomial coefficients make kernel evaluation impractical. `native_decide` trusts the compiler in addition to the kernel; both affected theorems carry an explicit trust annotation. No result in §§4–7 depends on them.
- *One named open goal remains*: the equality, for all n , of the recurrence-defined partner sequence with the formal square root of the bulk sequence (`open_goal_partner_eq_sqrt_s7`). It is the solution-level transport of the operator identity that Theorem 4.5 establishes at operator level, and it is the single remaining `sorry` in the development — carried as a named goal in `OpenGoals/` rather than as a silently weakened statement.

9.6. An honest assessment of the formalization. The proofs above are, as proofs, shallow: after the right statements are set up, almost all of them are

closed by `ring`. We do not claim otherwise, and we do not think it detracts from the contribution. What is difficult in this development is not discharging the goals but arriving at statements that are simultaneously (i) true uniformly in the template parameters rather than per candidate, (ii) division-free, so that they can be stated at all in the available library, (iii) non-vacuous, in the sense that they pin coefficients rather than assert existence, and (iv) honest about which of them rests on a citation. The partner of (9) being *determined* by the template rather than guessed, the collapse $\theta(P_2) = 2P_1$ making the identity polynomial, and the axiom quarantine are the three places where that work went. To our knowledge, no kernel-checked proof of a symmetric-square structure theorem for Picard–Fuchs-type operators has previously been given in any proof assistant.

10. REPRODUCIBILITY

Every number in this paper is produced by a script or a proof in one of two repositories, referred to below as **R1** (the formal development, Lean 4) and **R2** (the exact and numerical computations, Python with `sympy` and `mpmath`). Repository locations are given with the published version. Nothing in this paper is a constant typed from memory; the program’s standing rule is that a number may enter prose only by being read from an artifact that a reader can regenerate.

10.1. Artifact index.

Result	Status	Artifact
Thms. 4.5, 4.4, Cor. 4.7	(K)	Agora/Sequences/PartnerOperators.lean
Thm. 4.9	(K)	Agora/Swampland/SymSquareC3b.lean
Rem. 4.12	(K)	Agora/SymSquare.lean §3
Prop. 7.1	(K)	Agora/Sequences/FormalSqrt.lean
Thm. 7.2, Cor. 7.4, Rem. 7.5	(K)	Agora/Sequences/PartnerIntegrality.lean
Thm. 7.3 (as axiom)	(C)	Agora/Axioms/OBrien2016.lean, AXIOMS.md
Props. 3.1, 3.2, Rem. 3.3	(E)	check_L3_riemann_scheme.py → L3_RIEMANN_SCHEME.json
§5 in full	(E)	check_L3_irreducible_minimal.py → L3_IRREDUCIBLE.json
§5 controls	(E)	test_L3_irreducible_minimal_controls.py
Props. 7.7, Rem. 7.8	(E)	check_s7_partner_integrality_modular.py → S7_PARTNER_MOD
Prop. 7.10	(E)	check_s7_hauptmodul_gamma07plus.py → HAUPTMODUL_S7_GAMMA
NC 6.2, Prop. 6.3, NC 6.6	(N)/(E)	check_U1_lattice.py → C2_cooper_s7_v4.json
§6 controls	(E)	test_U1_controls.py
Prop. 6.4 (2nd implementation)	(E)	checkers/check_U1_splitting_independent.py
Prop. 6.4 controls	(E)	checkers/test_U1_splitting_independent_controls.py

The lattice certificate is content-addressed: the version underlying §6 has

SHA256 = 036cd895db892aee9802514ec18668535f8896cee53cf6123533af9844387c3e,

and the independent verification of Proposition 6.4 was run against that file and against its immediate predecessor, whose derived block is byte-identical, with the same result. Literature sources used in this paper were fetched and

hash-pinned before being read; the hashes are recorded in the repositories' provenance files.

10.2. Running the artifacts. R1 builds with `lake build`; the golden tests build with it. R2's checkers are standalone scripts, each taking an optional `--json` argument that writes its certificate, so a checker can be re-run and its output diffed against the committed certificate:

```
python3 checkers/check_L3_irreducible_minimal.py --json
/tmp/out.json
```

The independent splitting check takes the certificate path as an argument rather than embedding it, and reads the target level from the certificate at runtime:

```
python3 checkers/check_U1_splitting_independent.py --cert <path>
```

The control suites are runnable directly or under `pytest`.

10.3. Negative controls. The program's operating rule is that *a check which cannot fail is not a check*. Every computation above ships with controls designed to make it fail, and the controls are run, not merely written. The rule has a specific origin: an earlier computation in the program passed its own test because its statistic had been clamped below its own threshold, and the resulting claim had to be retracted. The controls exist so that this cannot recur silently.

Concretely, the controls used in this paper are:

- *Known-negative inputs.* The irreducibility checker is fed an operator with a hyperexponential solution by construction and one with distinct indicial roots at the origin; it must reject both, and does (§5).
- *Cross-checker agreement.* The L_2 exponents computed by one checker must reproduce, under pairwise summation, the L_3 Riemann scheme computed independently by another — at all four singular points, for both families.
- *Machinery controls.* The numerical continuation must recover a matrix known analytically before it is used on unknown loops (Remark 6.1).
- *Corruption controls.* Perturbing a recognized matrix must break a downstream exact gate; three distinct corruptions are exercised, failing at the involution gate, at the infinity-product gate, and by collapsing the invariant-form solution space to dimension 0 (§6). Corrupting a coefficient of the modular series must break the degree-2 verification (§7).
- *Discrimination controls.* The identical lattice pipeline run on the s_{10} family must produce a different, internally consistent answer, and does (NC 6.6); the modular fit tested against the wrong level must fail every fit, and does; the genuine s_7 Gram matrix tested against $U \oplus \langle 20 \rangle$ must fail, and does.

- *Vacuity controls.* A statement whose content is not obvious is accompanied by a check that it is not trivially true: the dyadic baseline of Proposition 7.1 is paired with a proof that $\frac{1}{3} \notin \mathbb{Z}[\frac{1}{2}]$.
- *Positive controls after corruption.* Every suite re-runs the unmodified input last, confirming that the scrambling hooks are surgical and leave no state behind.

Two further disciplines are worth stating because they constrain what the artifacts are permitted to emit. First, target values are assembled at runtime from data read out of the certificate under test, never hardcoded outside a labeled control — so a checker cannot accidentally test a constant against itself. Second, checkers that cannot derive a quantity emit `null` for it rather than a plausible value: the irreducibility certificate reports no rank at all, precisely because the step that would license one is a citation and not a computation (§8).

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